How do changes in housing supply, demand, and pricing affect property sales in US metropolitan areas?

[DS2500 – Team Project Final Report]

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1. **Introduction (Maverick)**
    a. **Goal of the Project**

The goal of this project is to analyze various aspects of the housing market, - including pricing, supply, and demand - using data from 2019 to 2025, to determine what factors most drive property sales within U.S. metropolitan areas.

   b. **Problem Statement & Background**

Our group decided to choose a topic related to housing because, as the housing crisis deepens and home prices continue to rise, many Americans are facing housing insecurity and homelessness. So, gaining insight into what most impacts sales could allow for a greater understanding of why there is a shortage of affordable housing, whether that be an issue of supply, demand, or affordability.

Thus, an analysis of the factors which motivate property sales could be beneficial for those seeking housing and toward understanding how housing may be affecting America's increased rates of homelessness. Additionally, such information would be especially useful to individuals and businesses in the fields of realty, construction, contracting, even politics - where campaigning and direct action could lead to great strides toward more accessible housing.

2. **Introduction to Our Data (Selen)**

    a. **Dataset Overview**

Source: Zillow Research Public Datasets (https://www.zillow.com/research/data/)

Time Period: 2019-2025

Size: 168; For-Sale Inventory, New_Listings, Newly_Pending_Listings, Median_Sale_Price, Mean_Sale_Price, Home_Value_Index, Median_Days_to_Close, Sales_Count

Format: CSV

b. Data Collection Methodology

Zillow Research Public Datasets is collected from public records, Multiple Listing Services (MLS), user contributions, and third party data providers such as the Census.

c. Key Variables

Variable Name	Description	Type	Example Values
For-Sale Inventory	Numerical value representing the number of for-sale home inventory in a given location	Numeric	73707
New_Listings	Number of new listings that have come on the market in a given month	Numeric	20008
Home_Value_Index	Measure of home value in a given region	Numeric	219565.59

d. Ethical Considerations

The ethical considerations for this project are largely its social implications. While the dataset doesn't contain any personal information, housing metrics can reflect social inequalities. Since the data represents metropolitan market behavior, it fails to capture the struggles of individuals facing housing insecurity, or acknowledge issues like homelessness or affordability. This creates the risk of misinterpretation by policymakers who may prioritize profit and overlook vulnerable populations. The analysis must be clear about its limitations to avoid this. Overall, the ethical responsibility of this project is in acknowledging the biases in the dataset and avoiding overgeneralization of the findings.

3. Data Science Approaches (Sahasra)

a. Data Preprocessing/Preparation

In order to prepare the data for analysis, we first had to clean it in order to get rid of any rows that could cause issues when running code. We first read the csv files and turned them into dataframes. Next, we selected columns from the dataframes that had values from 2024. After that, we cleaned the data in each data frame by checking each row for 0 or null values in order to eliminate them. Finally, we redefined the data frame with only the shared columns across all the data frames. We ended up one cleaned and shared data frame with data from 2024 from all 8 csv files.

b. Exploratory Data Analysis

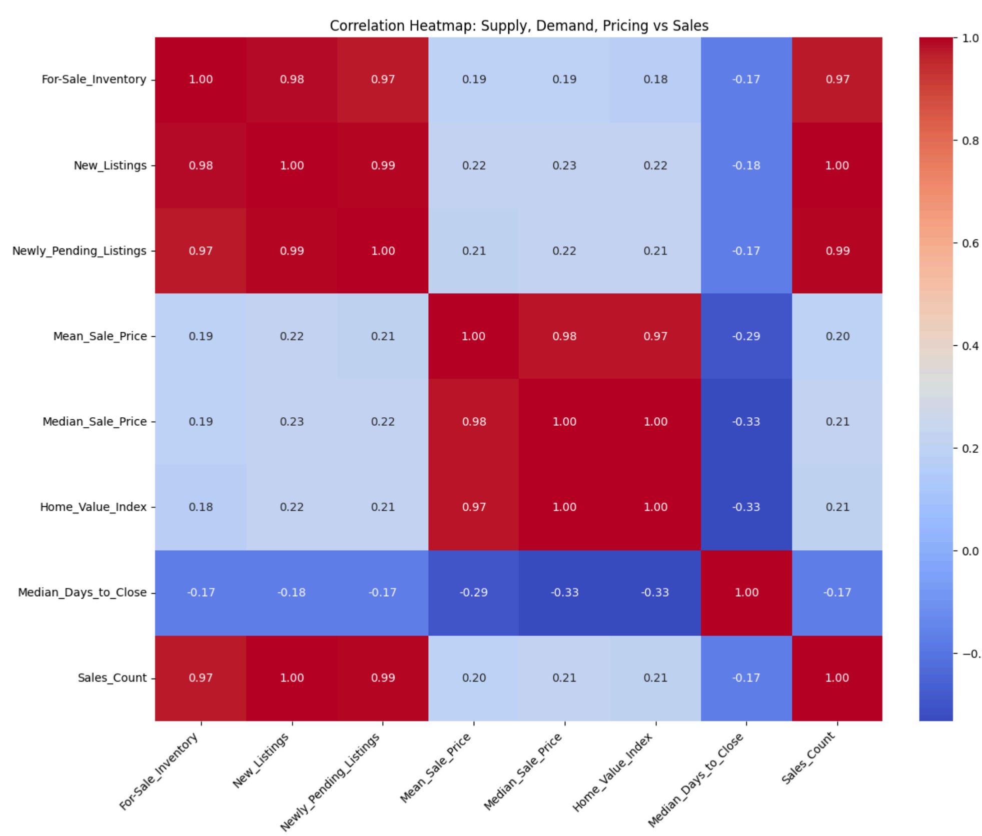
First, we created a correlation matrix to find weak and strong correlations between numerical features, excluding things like Region ID. We noted the strongest correlation and weakest correlation to Sales Count for later analysis.We then performed linear regression and logistic regression in order to understand the relationship between the variables more in order to hopefully predict future possible data accurately. Finally, we performed KNN regression and KNN classification to check our accuracy and precision of the regression models.

c. Modeling and Evaluation

After making our correlation matrix, we made a heatmap and color coded it in order to easily show where the areas of strongest and weakest correlation lay between variables. We also modeled a confusion matrix after performing KNN classification to check how right or wrong our code was. Finally, we made several scatter plots to show different relationships: Demand vs. Sales Count (Newly Pending Listings), Supply vs. Sales Count (For-Sale Inventory), Pricing vs. Sales Count (Median Sale Price), and Speed of Sales vs. Sales Count.

4. Results and Conclusions (Marie)

a. Model Performance

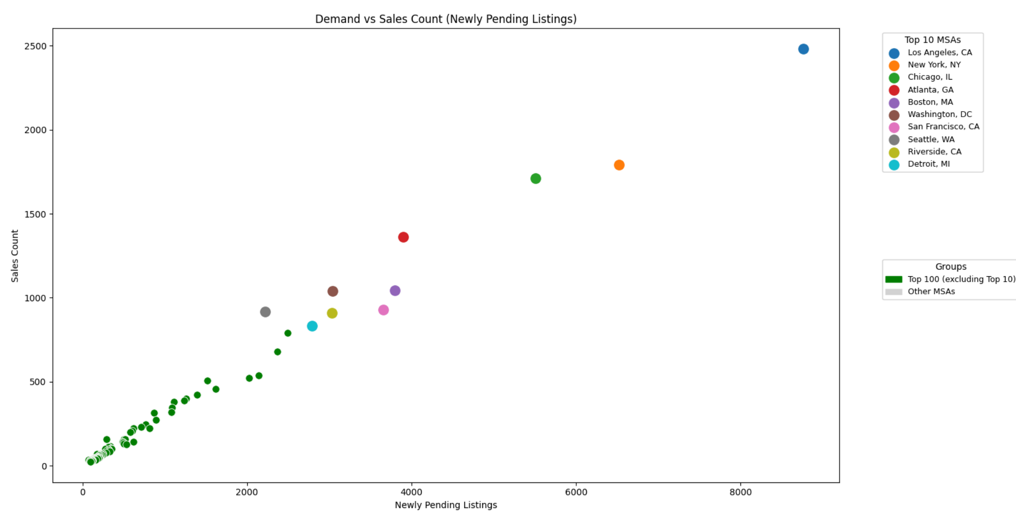


The correlation heatmap provides insight into how numerical features interact with each other and with **Sales_Count**:

- **New_Listings shows the strongest correlation with sales (1.00).** This highlights the direct effect of greater supply leading to increased sales activity. For-Sale_Inventory (0.97) and Newly_Pending_Listings (0.99) also show strong correlations, reinforcing the central role of supply and demand throughput.
- **Median_Days_to_Close has the weakest correlation with sales (−0.17).** This suggests that transaction speed is not a significant determinant of whether a sale occurs, though it may reflect market efficiency.
- **Pricing metrics (Mean_Sale_Price and Median_Sale_Price) are nearly identical in their correlations.** Both track closely with each other and with the Home_Value_Index, indicating internal consistency. Their similarity may suggest a relatively normal distribution of sale prices across Metropolitan Statistical Areas (MSAs).

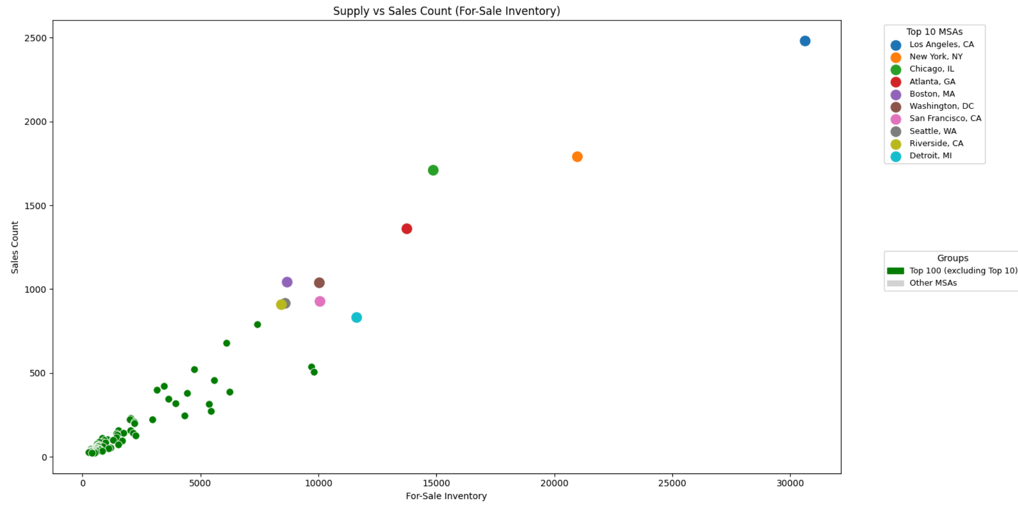
b. Key Findings

i. The Influence of Buyer Demand on Sales Volume



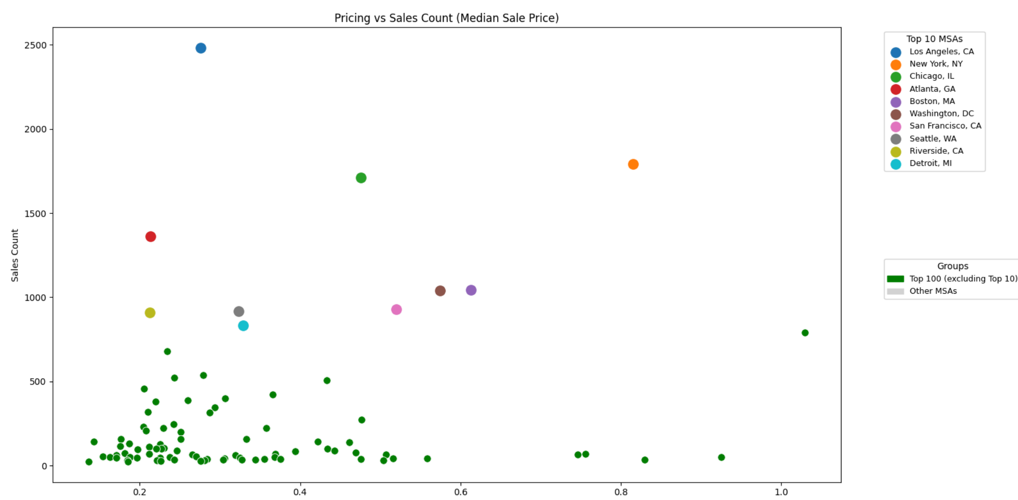
Property sales are primarily driven by buyer demand, not pricing or transaction speed. Metropolitan areas with higher numbers of newly pending listings consistently record more completed sales, showing a near-linear relationship between contracts signed and sales counts. This pattern holds across both large and small MSAs, but the effect is most pronounced in top markets like Los Angeles and New York, where demand volume translates almost immediately into higher sales throughput.

ii. The Role of Market Inventory in Determining Transaction Activity



Sales activity is tightly linked to the amount of inventory available on the market. Metropolitan areas with larger for-sale inventories consistently achieve higher sales counts, confirming that supply acts as a throughput constraint. Without sufficient listings, even strong demand cannot convert into transactions. Top MSAs cluster at the high end of both inventory and sales, underscoring how scale amplifies the supply–sales relationship.

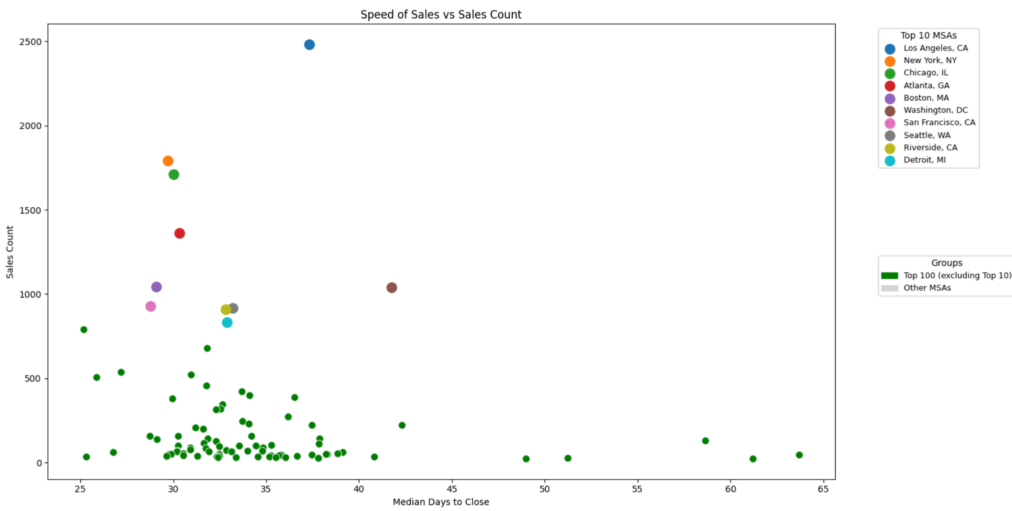
iii. The Limited Effect of Pricing on Sales Outcomes



Median sale price does not significantly determine sales volume across metropolitan areas. While pricing metrics such as mean and median sale price are internally consistent and strongly correlated with each other, their relationship with sales count is weak. High-price markets like San Francisco and New York still record strong sales, while lower-price markets do not necessarily outperform.

This suggests that pricing reflects valuation trends rather than driving transaction volume.

iv. The Marginal Impact of Transaction Speed on Market Throughput



Markets with shorter median days to close tend to record slightly higher sales counts, but the effect is secondary compared to supply and demand. The weak negative correlation indicates that transaction speed reflects efficiency and liquidity rather than being a decisive factor in whether sales occur. Top MSAs generally combine higher sales with faster closings, but across the broader dataset, speed alone does not explain differences in sales activity.

c. Conclusion

The analysis demonstrates that housing supply and demand are the primary determinants of property sales in U.S. metropolitan areas. New listings and pending contracts show near-perfect correlations with sales counts, confirming that transaction volume rises almost directly with market throughput. Pricing metrics, while internally consistent, exert only a limited influence on sales activity, and transaction speed contributes marginally, reflecting efficiency rather than scale.

City size further amplifies these dynamics, with larger metropolitan areas consistently converting supply and demand into higher sales volumes. Overall, strategies that expand inventory and strengthen buyer engagement will have the greatest impact on sales efficiency, while pricing and transaction speed should be monitored for affordability and liquidity but remain secondary to the fundamental drivers of supply and demand.

Taken together, these findings answer the central question: *changes in housing supply and demand directly drive property sales in U.S. metropolitan areas, while pricing* *and transaction speed play secondary roles.* For policymakers, investors, and market participants, this suggests that strategies to increase listing volume and buyer engagement will have the greatest impact on sales activity, while pricing adjustments should be used to optimize margins and efficiency rather than to influence transaction volume.

5. Future Work & Limitations (Maverick)

a. Limitation of Current Analysis & Future Research Directions

There were limitations in this project, not only in the reduced volume of data after removing empty rows, but also in the features we chose to analyze. Our group chose various aspects of pricing, supply, and demand - features which are interdependent in the housing market. Thus, future work should include aspects with a more indirect correlation to housing sales such as data on average household size and income, education level, race and ethnicity, and ages of the individuals purchasing homes. This way an analysis of what motivates property sales may be conducted on the basis of how people's personal circumstances and demographics influence their ability to make property purchases and therefore influence the housing market. Further, our data focuses on metropolitan areas in particular, which are often more populous and have more expensive housing. So, it could prove beneficial to analyze other types of regions within the United States, including rural, urban, and suburban areas. This additional data could be analyzed independently or collectively to understand how sales vary amongst these different regions.

References:

[Housing Data. 2006-2025. Zillow. https://www.zillow.com/research/data/]